Cross-Semantic Heterogeneous Modeling Network for Hyperspectral Image Classification

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***Abstract*— The adequate and finer spectral information in hyperspectral images (HSIs) are benefit for various downstream applications like smart agriculture and environmental monitoring. In HSI classification, dual-stream convolutional networks have gained much attention and have been widely used. In patch-based hyperspectral classification tasks, however, merely using center-labeled patches could lead to an increased unlabeled noise in the data. Moreover, in the application of dual-stream network structures, heterogeneity existed in both the data and feature semantic levels to capture more representative features. To tackle these challenges, we have devised a framework called cross-semantic heterogeneous modeling network (CreatingNet), which aligns more closely with the design principles of dual-stream networks by adjusting the input size. This framework introduces a distance metric attention mechanism (DMAM) based on spectral and spatial distances to strengthen the influence of the center pixel on the entire patch. Additionally, we present a fusion module named CrossViT, which combines features with diverse structures and characteristics, leveraging their complementarity. The proposed multiscale heterogeneous fusion module allows for more effective integration of spatial and spectral features in the images. Extensive experiments on four well-known HSI datasets (Indian Pines, Pavia University, Salinas, and Houston 2013) demonstrate the superior classification performance of the proposed CreatingNet to several state-of-the-art methods. The effectiveness of the proposed model is further validated through ablation studies.**

***Index Terms*— Attention mechanism, fusion, hyperspectral imagery (HSI), image classification, spectral–spatial feature extraction.**

I. Introduction

HYPERSPECTRAL image (HSI) consists of narrow and contiguous spectral bands within the electromagnetic spectrum, encapsulating a wealth of both spectral and spatial information. The rapid advancement of hyperspectral technology has propelled Earth observation activities to new heights. Hyperspectral technology finds applications in diverse fields, including medical diagnosis, food and drug testing, smart agriculture, and environmental monitoring. Hyperspectral technology has been adopted in several real-world applications, including but not limited to the atmosphere, agriculture, mineral exploration, and coastal zones [1]. When compared to multispectral data, HSI enables the detection and identification of substances on Earth with a finer and more accurate granularity. In the field of intelligent HSI feature extraction, numerous practical applications have emerged, such as unmixing [2], [3], image super-resolution [4], [5], and target detection [6], [7]. These directions show great potential in leveraging the abundant information provided by HSI to tackle real-world problems and challenges. HSI classification is a research domain with significant practical importance, aiming to assign a unique label to each pixel in an image. Nevertheless, it encounters several challenges, including spectral variability, intricate noise interference, high dimensionality, limited sample size, and data redundancy.

In recent years, numerous studies have been undertaken to enhance the classification performance of HSI. Traditional HSI classification typically extracts features from HSI using pattern recognition methods, and then followed by an interpretable classifier [8], [9]. Well-established examples of this statistical feature extraction approach encompass techniques such as linear discriminant analysis (LDA) [10], local preserving discriminant analysis (LFDA) [11], sparse preserving projection (SPP) [12], among others. Apart from being treated as feature sequences, HSI also exhibits image-like characteristics. In response, researchers have introduced various methodologies that integrate spatial and spectral features, such as extended mathematical contour (EMP), extended attribute profiles (EAPs), morphological profiles (MPs) [13], gray co-occurrence matrix (GCM) [14], Gabor filter (GF) [15], and others. Leveraging sparse representation and classification methods have demonstrated promising results [16], [17]. Though effective in HSI classification, traditional machine learning methods may lack robustness in data fitting and feature representation capabilities [16], [18]. The advent of deep learning has brought forth significant progress in HSI classification, harnessing the potential of deep neural networks.

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HSI offers two prominent avenues for feature extraction. Its intrinsic image-like property enables the extraction of abundant spatial feature information. Furthermore, each HSI pixel corresponds to a spectrum representing the reflectance of specific wavelengths for distinct materials, thereby presenting an opportunity for extracting spectral feature information. Historically, some researchers have introduced techniques based on spectral feature extraction. However, these techniques tend to overlook the spatial characteristics of HSI. Conversely, others have advocated for methods based on spatial feature extraction, inadvertently neglecting the comprehensive modeling of HSI's spectral features. Consequently, the pursuit of a network structure capable of delving deep into both spatial and spectral features of HSI has become a prevailing approach in HSI classification within the domain of deep learning [19], [20], [21]. Zhou et al. [22] introduced a novel multiple feature fusion model termed attention multihop graph and multiscale convolutional fusion network, which includes two subnetworks of multiscale fully CNN and multihop GCN to extract the multilevel information of HSI. Duan et al. [23] build a multiple neighborhood reconstruction model to reveal the essential resemblance of multiple pixels in nonlinear spectral feature space. Guo et al. [24] propose a dual-view spectral and global spatial feature fusion network to extract the spatial–spectral features for HSI classification, including a spatial subnetwork and a spectral subnetwork. Duan et al. [25] propose a novel multiple neighborhood-aware nonlinear collaborative analysis method. Relying on the primary and secondary neighborhoods, a dual-level neighborhood reconstruction is designed to search for optimal neighbors and mine the common attributes within the neighborhood [25]. Jia et al. [26] proposed a center-masked transformer approach for improving the performance of transformer-based networks in the HSI classification task with limited samples.

For simultaneously incorporating spectral and spatial features from HSI, diverse approaches have been explored. For example, Chen et al. [27] introduced a 3-D-convolutional neural network (CNN) structure to enhance HSI classification, while Roy et al. [28] combined 3D CNN and 2D CNN through a concatenation. Ahmad et al. [29] integrated fundamental concepts from both 3-D and 2-D initial networks along with attention. Sun et al. [30] devised a unified model called spectral–spatial feature tokenization transformer, which melds 3-D and 2-D convolutions with Transformer architectures. This clever design accommodates both spatial and spectral aspects. However, it is important to note that for image data, the spectral and spatial dimensions differ in their physical meanings. Equalizing the two stride levels overlooks the physical significance of the spectral dimension, impairing interpretability and substantially increasing the number of parameters compared to 2D CNN. This, in turn, exacerbates the challenges concerning training time and storage consumption.

In recent years, transformer architecture has risen to prominence for its ability to capture long dependencies within word sequences, establishing itself as a robust structure. Its application in the field of HSI classification is noteworthy. Hong et al. [31] were the trailblazers in introducing the transformer structure and unveiled the SpectralFormer model. He et al. [32] adopted a sequential strategy by using a CNN to extract spatial features and a transformer with dense connections to capture spectral relationships. Huang et al. [33] incorporated contrastive learning and tailored the swin transformer to the 3-D nature of HSI. Yang et al. [34] combined CNN and the transformer, and created FusionNet to fully harness HSI features. Additionally, Ibanez et al. [35] developed masked autoencoding and achieved remarkable outcomes. These methods exemplify diverse approaches to integrate the transformer architecture into HSI classification, underscoring its efficacy in leveraging both spectral and spatial information.

Another approach for joint modeling spectral and spatial features is to use a dual-stream network structure to separately extract spatial and spectral features, followed by a feature fusion [36]. Many researchers have explored this perspective as well. Li et al. [21] used spatial attention and spectral attention to extract features, creating a double-branch dual-attention mechanism network. Xue et al. [37] introduced a residual structure on the basis of these two attention mechanisms. Hang et al. [38] designed a two-branch spectral–spatial attention network and introduced spectral and spatial attention mechanisms separately. Hong et al. [39] incorporated graph convolutional structures into HSI classification and developed a simplified version of this method. In terms of graph networks, Yao et al. [40] proposed the deep hybrid multigraph neural network, which also showed promising results. However, the original intention of using a dual-stream network is to have more easily learnable structure for different branches, allowing the network to smoothly learn representative features, which is often overlooked.

In patch-based classification methods, networks typically treat each patch as a unified entity. In HSI classification, although the class label of a patch is contingent upon the label associated with the spectrum of its central pixel, the surrounding pixels also supply potentially valuable spatial information for effective discrimination. Liu et al. [41] established a guiding principle that underscores the pivotal role of the center pixel within a patch and introduced a central attention mechanism for HSI classification. When treating the entire patch as a single input, different spectral bands remain independent, consequently disregarding the intrinsic sequential structure within HSI.

Moreover, in HSI classification, the issue frequently revolves around the scarcity of data [42], [43]. The limited size of the training dataset means that excessively intricate network architectures often succumb to overfitting [44]. However, simple network structures lack the capacity to extract precise features from the copious but repetitive data. Furthermore, it is imperative to employ the feature fusion strategy to extract multiangular features from images [45], [46]. Cui et al. [47] introduced the multiscale spatial–spectral convolutional network with image-based framework, which incorporated dilated convolutions and fusion strategies, and demonstrated impressive results. Yang et al. [48] devised an

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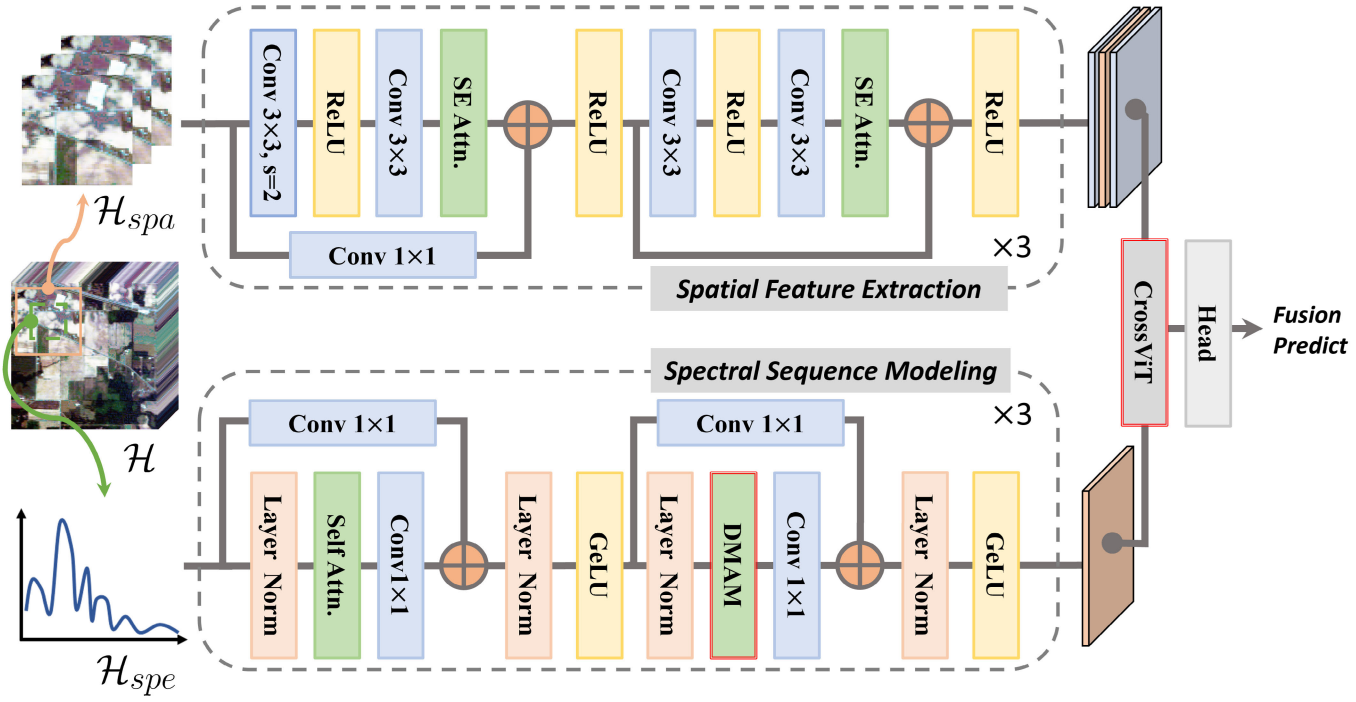


Fig. 1. Overall framework of the proposed CreatingNet for the HSI classification.

enhanced multiscale feature fusion network, emphasizing a multiscale feature fusion strategy in their approach. In most current research, employing a specialized and efficient network structure is necessary [49], [50], [51].

In practical scenarios, it is important to note that only the spectral values of the central or target pixel directly correspond to land-cover classes. When working with larger patches, it becomes evident that many of the surrounding pixels often have varying classes compared to the central pixel, thereby introducing a considerable amount of noise in the spectral information utilized for land-cover discrimination. We refer to this noise as unlabeled noise.

Nowadays, how to efficiently mine the spatial–spectral information from images is the core problem that needs to be solved in HSI classification tasks. When using a dual-stream network structure to extract spatial–spectral information separately, heterogeneity arises at the semantic levels of both data and features, enhancing the ability to effectively distinguish between the two aspects of information. Furthermore, this is manifested in different raw input feature dimensions and extracted feature dimensions.

To address the aforementioned issues, the core contributions of our proposed cross-semantic heterogeneous modeling network (CreatingNet) are summarized as threefold.

1) To tackle the challenge of varying spectral features in HSI classification, we propose a dual-stream network structure called CreatingNet. To more effectively align with the design principles of the dual-stream network, we have implemented a specialized design at the input layer aimed at enhancing the complementarity between spatial and spectral features.
2) To address the issue of unlabeled noise in patch-based HSI classification, we propose a novel attention mechanism called the distance metric attention module (DMAM). When used in network feature extraction, this module can reduce feature noise by leveraging the correlation information among spectral bands, thereby enhancing the effectiveness of feature extraction.
3) To address the heterogeneity between spectral and spatial features, we introduce CrossViT, a cross-modal feature fusion module. This module utilizes the inherent characteristics of 3-D hyperspectral images (HSIs) in both the spatial and 2-D spectral domains. By merging feature descriptors, CrossViT effectively exploits the complementarity between the two feature sets, thereby enhancing the model's overall discriminative capability.

II. Proposed Method

Considering the challenges outlined in HSI classification, our proposed approach, CreatingNet, depicted in Fig. 1, comprises the heterogeneous input module, spatial feature extraction, spectral sequence modeling, and CrossViT fusion module.

A. Heterogeneous Input Based on Different Size

Historically, the concept of employing a dual-stream network architecture for HSI classification primarily revolved around distinct network structure designs. One branch was engineered for the extraction of spatial information features, while the other was dedicated to extracting spectral information features.

When employing CNN architectures for the extraction of spatial information features, the fundamental idea is

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to treat HSI as a stack of various feature patches. The contribution from neighboring pixels to the target pixel in the spatial dimension primarily manifests in semantic attributes at multiple levels, including texture structure and scene description. In this context, larger patches are advantageous in providing richer information, thus enabling the extraction of more effective features.

In contrast to conventional computer vision tasks, HSI is unique in its possession of continuous sequential spectral information, which stands as the most crucial discriminative factor in land use and land-cover (LULC) classification. In the natural world, materials consist of molecules or atoms, each with its own unique molecular or atomic structure. Stemming from Planck's hypothesis of energy quanta, the absorption and emission of energy by these atoms and molecules occur discretely, not continuously. Hence, individual atoms and molecules exhibit distinct reflectance spectra. Furthermore, materials are formed from different combinations of atoms and molecules, which results in their characteristic spectral patterns in terms of reflectance. Therefore, the effective extraction and utilization of spectral features stand as a pivotal concern in HSI classification.

However, there is a problem with unlabeled noise in patch-based HSI classification. To mitigate this issue, a prevalent strategy in current research involves assigning distinct initial weights to pixels based on their proximity to the central pixel. This is undertaken to reduce the influence of non-central pixels on the classification

It is essential to emphasize that, in addition to having an extensive array of spectral bands, HSI provides a genuine, overhead representation of geographical information, setting it apart from conventional images. Adhering to Tobler's First Law of Geography asserts that "Everything is related to everything else, but near things are more related to each other." When applied to HSI, it suggests that within the immediate vicinity of a target pixel, there is a substantial likelihood of encountering pixels belonging to the same class. These neighboring pixels often exhibit similar spectral characteristics. Consequently, when we evaluate the spectral sequences of individual pixels in isolation, opting for smaller patches yields a higher unlabeled noise. Motivated by this, we adjust the input structure and input size of the two branches. These adjustments are aimed at ensuring a more seamless alignment with the underlying network structures of each branch, considering their distinct mechanisms.

Given an HSI ($\mathcal{H} \in \mathbb{R}^{C\times N_r\times N_c}$), with N_r and N_c denoting spatial length and width, and C representing spectral channels, the class distribution of a single pixel (i, j) is defined as $p(i, j)$ with its actual value denoted as $y_{\text{true}}(i, j)$. To incorporate neighborhood information, we select a patch of size $w = 2s + 1$, centered at coordinates (i, j) as input, where s is the stride. This patch undergoes a transformation via a shape transformation function θ as a structured input for the network, represented as

$$\mathcal{H}_\tau(i, j) = \theta_\tau(\mathcal{H}, w_\tau, i, j) \tag{1}$$

where subscript τ denotes the representation in either the spatial or spectral dimension, with $\tau \in \{\text{spa}, \text{spe}\}$ representing spatial and spectral feature extraction streams, respectively. Note that in a typical dual-stream network structure, θ is completely identical for spa and spe. However, the length of w in (1) is different between the two streams, with $w_{\text{spa}} > w_{\text{spe}}$.

For the spatial feature extraction stream, the data is integrated into a form suitable for CNN input. Specifically, after patch cropping, $\mathcal{H}$ becomes a 3-D tensor of size $\mathcal{H}_{\text{spa}} \in \mathbb{R}^{C\times w_{\text{spa}}\times w_{\text{spa}}}$. The data can be described as C feature maps, each of size $w_{\text{spa}} \times w_{\text{spa}}$.

For the spectral feature extraction stream, in order to utilize the spectral information from the neighborhood, $\mathcal{H}_{\text{spe}}$ is cropped to a 3-D tensor of size $\mathcal{H}_{\text{spe}} \in \mathbb{R}^{C\times w_{\text{spe}}\times w_{\text{spe}}}$ and then flattened spatially. The data is transformed into a 3-D matrix of size $\mathcal{H}_{\text{spe}} \in \mathbb{R}^{C\times w_{\text{spe}}^2}$.

Adopting this strategy, we have established distinct input structures for the two branches within the dual-stream network, bringing them into closer alignment with the network's design principles and its individual modules. Detailed descriptions of each module used in the whole framework of CreatingNet are presented in the following sections.

B. Spatial Feature Extraction Based on Residual Attention Network

CNN architectures have consistently showcased their remarkable feature representation capabilities across a spectrum of computer vision tasks. In the context of HSI classification, the application of CNNs for spatial feature extraction is a prevalent practice. Nevertheless, it is important to recognize that HSI classification, particularly when employing patch-based methods, deviates from typical computer vision tasks in terms of image size and data volume. Conventional CNN structures frequently involve a substantial number of parameters, which may not be suitable for HSI applications if applied directly.

Numerous prior works have employed modest and uncomplicated CNN architectures within dual-stream networks to address the limited data volume characteristic of HSI. Nevertheless, these shallow network designs might not adequately capture the profound features of HSI. To resolve this disparity, this article enhances the latter portion of the ResNet18 network, employing it as the foundational architecture.

This backbone network offers two key advantages in the context of HSI classification. First, as opposed to the typical CNN processing in HSI classification, the incorporation of skip connections within the network effectively mitigates the issues associated with gradient vanishing and explosion in deep networks, thereby enhancing network performance. Second, instead of relying on pooling layers for downsampling, we achieve downsampling by increasing the stride of convolutional kernels. While traditional convolution and pooling operations quickly reduce the size of the network feature maps and expand the receptive field, it is important to note that the patch size in HSI classification is usually small. In such cases, direct application of pooling frequently results in excessive reduction of feature maps, resulting in a shallow network depth. Conversely, increasing the convolutional kernel stride yields a more progressive downsampling process.

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Given that HSI possesses a larger number of feature maps in comparison to natural images, the incorporation of channel attention mechanisms at later stages can significantly enhance the network's capacity to learn the most informative channels within the feature maps.

The squeeze-and-excitation (SE) layer represents (the SE Attn. module of the spatial stream in Fig. 1) a lightweight channel attention model, which serves to amplify the significance of pivotal channels. Thanks to its streamlined design, integrating this attention mechanism does not predispose the model to overfitting.

In the squeeze (SQ) phase, the input features $x = [x(c, i, j)]_{C\times w_r\times w_c} \in \mathbb{R}^{C\times w_r\times w_c}$ are compressed into a vector of the same length as the number of channels. This compression is achieved using global average pooling, which compresses the values in the $w_c \times w_r$ spatial dimensions and outputs a vector of length C. Specifically, for the data $x(c, :, :)$ in channel c

$$f_{\mathrm{SQ}}(x(C, :, :)) = \frac{1}{w_r \times w_c}\sum_{i=1}^{w_r}\sum_{j=1}^{w_c} x(C, i, j). \tag{2}$$

In the excitation phase, the interdependencies among channels are captured through the gating mechanism of two nonlinear fully connected layers

$$F_{\mathrm{SE}}(x) = \mathrm{Sigmoid}(\boldsymbol{W}_2\mathrm{ReLU}(\boldsymbol{W}_1 f_{\mathrm{SQ}}(x)) \tag{3}$$

where $\boldsymbol{W}_1 \in \mathbb{R}^{(C/16)\times C}$ and $\boldsymbol{W}_2 \in \mathbb{R}^{1\times(C/16)}$. The channel attention module assigns adaptive weights to different channels, capturing their relative importance.

Taking inspiration from the original version, we have integrated this module after each block to bolster channel attention across the entire network structure. Essentially, the SE layer incorporates two fully connected layers to facilitate channel-specific processing and weight learning for various regions within the feature map, thus highlighting the more informative areas.

Through the above procedures, we have devised a fundamental unit structure named "spa-bottleneck" for extracting spatial information features.

For a network input x, we first pass it through a feature extraction block consisting of a two-stride 3×3 convolutional layer, a ReLU activation layer, a 3×3 convolutional layer, and an SE layer, resulting in $f_1(x)$. For the same input x, we perform dimensionality reduction by applying a 1×1 convolutional layer, yielding $f_d(x)$. We then add $f_1(x)$ and $f_d(x)$ through skip connections and apply a ReLU activation function, as follows:

$$u_{\mathrm{spa}}(x) = \mathrm{ReLU}(f_1(x) + f_d(x)). \tag{4}$$

Similarly, the second block follows a similar structure to the first block, but without downsampling and dimensionality reduction. For the input u_{spa}, we apply a 3×3 convolutional layer, a ReLU activation layer, a 3×3 convolutional layer, and an SE block in the feature extraction block, resulting in $f_2(u_{\mathrm{spa}})$. Since no dimension reduction is needed in this case, we directly add $f_2(u_{\mathrm{spa}})$ to u_{spa}. Finally, we apply a ReLU activation function. The overall process can be described by the following formula:

$$F_{\mathrm{spa}}(x) = \mathrm{ReLU}(f_2(u_{\mathrm{spa}}(x)) + u_{\mathrm{spa}}(x)). \tag{5}$$

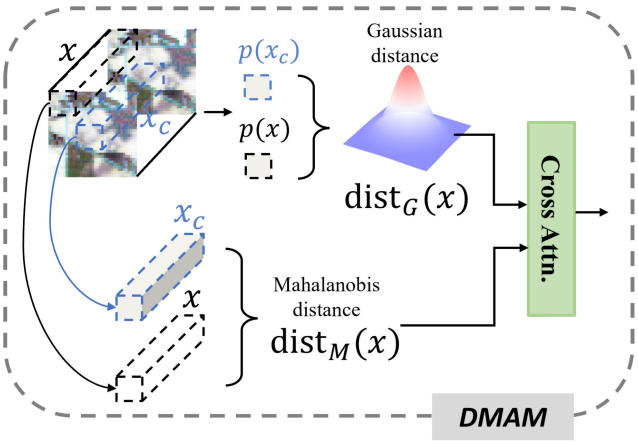


Fig. 2. Proposed distance measures attention module framework.

C. DMAM: Distance Measures Attention Module

In the extraction of spectral feature information, as previously mentioned, we treat the spectra within a patch as uniform inputs. A spectral sequence can be considered as 1-D data with contextual connections, bearing a resemblance to sentences in the field of natural language processing (NLP). In this analogy, each energy value, reflecting a specific electromagnetic wavelength, can be seen as an encoding of a word.

Notably in NLP, the transformer has emerged as a potent structure in recent years, well-suited for capturing long-range dependencies among words. Within the visual domain, the vision transformer (ViT) has demonstrated its remarkable capabilities across a variety of visual tasks. However, it is important to highlight that HSI, unlike general visual tasks, still encompasses sequential information along the spectral dimension. This distinctive characteristic contradicts the core design concept of the ViT.

In this article, we refine the multihead self-attention (MSA) mechanism, which is a fundamental building block of the transformer. Instead of viewing HSI classification as a traditional visual task, we draw direct inspiration from NLP. In contrast to ViT, our approach circumvents tokenization and position encoding operations and treats each spectral sequence as an individual token.

The effectiveness of the transformer crucially hinges on the application of MSA, a technique that involves stacking multiple self-attention mechanisms. As the name implies, self-attention excels at capturing internal correlations within data or features, thus diminishing the reliance on external information. Now, we give a brief introduction to self-attention and cross-attention mechanisms.

The self-attention mechanism is essentially aimed at enabling the network model to attend to correlations among different parts of the entire input. For each input vector x of length m, where each feature embedding vector is

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denoted as x_i, we multiply each embedding x_i by three transformation matrices $\boldsymbol{W}_q$, $\boldsymbol{W}_k$, and $\boldsymbol{W}_v$, resulting in query $\boldsymbol{Q} = [\boldsymbol{q}_1, \ldots, \boldsymbol{q}_m]^{\mathrm{T}}$, key $\boldsymbol{K} = [\boldsymbol{k}_1, \ldots, \boldsymbol{k}_m]^{\mathrm{T}}$, and value $\boldsymbol{V} = [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m]^{\mathrm{T}}$.

Next, the attention scores between each Q vector and each K vector are calculated in the form of dot products, $u_{i,j} = \boldsymbol{q}_i^{\mathrm{T}} \cdot \boldsymbol{k}_j / \sqrt{d}$, where d is the dimensionality of $\boldsymbol{q}_i$ and $\boldsymbol{k}_j$. After applying the Softmax activation function across columns, the attention scores are multiplied with a matrix $\boldsymbol{V}$ to generate the attention representation

$$F_S(x) = \mathrm{Softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^{\mathrm{T}}}{\sqrt{d}}\right)\boldsymbol{V}. \tag{6}$$

The cross-attention mechanism is an information interaction technique that calculates the correlation between two distinct modalities of data. This mechanism receives two inputs, x_1 and x_2. In contrast to the self-attention mechanism, the query vectors originate from x_1, while the key and value vectors originate from x_2. The subsequent computational process maintains consistency with the self-attention mechanism, i.e.,

$$F_C(x_1, x_2) = \mathrm{Softmax}\left(\frac{\boldsymbol{Q}_{x_1}\boldsymbol{K}_{x_2}^{\mathrm{T}}}{\sqrt{d}}\right)\boldsymbol{V}_{x_2}. \tag{7}$$

When utilizing sequential spectral information within patches, we aim to filter out effective information by focusing more on spectra that are more similar to the central pixel. This can reduce the impact of noise on the results. Therefore, a distance-based attention mechanism is designed, measuring the distances between various spectra at both spatial and spectral levels.

Unlike the common assumption of bands being relatively independent, HSIs have an exceptionally narrow wavelength range. Consequently, each spectral vector can objectively form a complete and continuous spectral curve. As a result, there is a certain degree of correlation between various bands in HSI. For this reason, we employ Mahalanobis distance to measure the band similarity between a pixel $\boldsymbol{x}$ and the center $\boldsymbol{x}_c$ (Fig. 2)

$$\mathrm{dist}_M(\boldsymbol{x}) = \left[(\boldsymbol{x} - \boldsymbol{x}_c)^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x} - \boldsymbol{x}_c)\right]^{-0.5}. \tag{8}$$

For any given input patch, a uniquely labeled center pixel can always be obtained. Here, we define $\boldsymbol{x}_c$ as the spectral vector of the center pixel. $\boldsymbol{\Sigma}$ represents the covariance matrix of the input.

In spatial terms, as the spatial locations between pixels can be approximately mapped to their relative locations in the geographical space in reality, we use Gaussian distribution surfaces as an approximate substitute (Fig. 2)

$$\mathrm{dist}_G(\boldsymbol{x}) = \exp\left(-\frac{(p(\boldsymbol{x}) - p(\boldsymbol{x}_c))^2}{2}\right). \tag{9}$$

The spatial relative location coordinates of the spectral vector x for a certain input are defined as $p(x)$, where p is a mapping from the spectral vector to spatial coordinates.

As a result, the proposed distance measures attention module (DMAM) can be represented as

$$F_D(x) = F_C(\mathrm{dist}_G(x), \mathrm{dist}_M(x)) \tag{10}$$

where x is the spatially flattened input; dist_M and dist_G operates pixel-wisely.

This architecture, originating from the fact that only the center pixel of a hyperspectral patch is annotated, devises two distinct distance metrics for spatial and spectral distances to serve as the core drivers for the attention mechanism, as shown in Fig. 2. The cross-attention mechanism facilitates efficient information exchange between these two distance metrics, forming a specialized attention mechanism tailored for hyperspectral data. This framework will be incorporated into the spectral sequence modeling module outlined in the next section.

D. Spectral Sequence Modeling Based on Mixed Attention Mechanism

In spectral sequence modeling, we construct the model using the previously mentioned self-attention and DMAM. Utilizing this hybrid spectral sequence modeling attention model enables more effective extraction of meaningful features from the spectral data of the image. The specific operations are as follows.

For an input x, we first apply layer normalization to it, which acts as the input to the self-attention mechanism. Subsequently, feature embedding is performed through a fully connected layer. To ensure dimensionality consistency, we apply a fully connected layer to the original input as well, and then add it to the result obtained from the attention mechanism. Finally, layer normalization and the GELU activation function are applied to this feature embedding. This process can be represented by

$$u_m(x) = \mathrm{GELU}(\boldsymbol{W}_1 F_S(x) + \boldsymbol{W}_2 x) \tag{11}$$

where $\boldsymbol{W}_1$ and $\boldsymbol{W}_2$ are the weights of the fully connected layers. Then, we append a similar residual DMAM block to this structure, substituting the self-attention mechanism with the previously mentioned DMAM. This can be expressed as follows:

$$F_{\mathrm{spe}}(u_m(x)) = \mathrm{GELU}(\boldsymbol{W}_1 F_D(u_m(x)) + \boldsymbol{W}_2 u_m(x)) \tag{12}$$

where $\boldsymbol{W}_1$ and $\boldsymbol{W}_2$ denote the weights of the fully connected layers.

E. CrossViT: Heterogeneous Feature Fusion Module

In the context of joint land-cover classification utilizing the spatial and spectral features of HSI, we recognize the importance of effectively leveraging both the spatial and spectral feature information. Simply overlaying or adding the two modes directly may obscure their distinctive characteristics, potentially limiting the classification performance.

As shown in Fig. 3, within our network architecture, the 3-D feature map extracted from spatial feature extraction is $H_{\mathrm{spa}} \in \mathbb{R}^{512 \times w_0 \times w_0}$. The 2-D feature map obtained from spectral sequence modeling is $H_{\mathrm{spe}} \in \mathbb{R}^{4 \times 512}$. To ensure that features from spatial and spectral can be aligned, we first flatten the 3-D feature map to ensure a uniform dimensionality for both feature maps. Furthermore, we adjust the spatial feature size through two fully connected layers to align with

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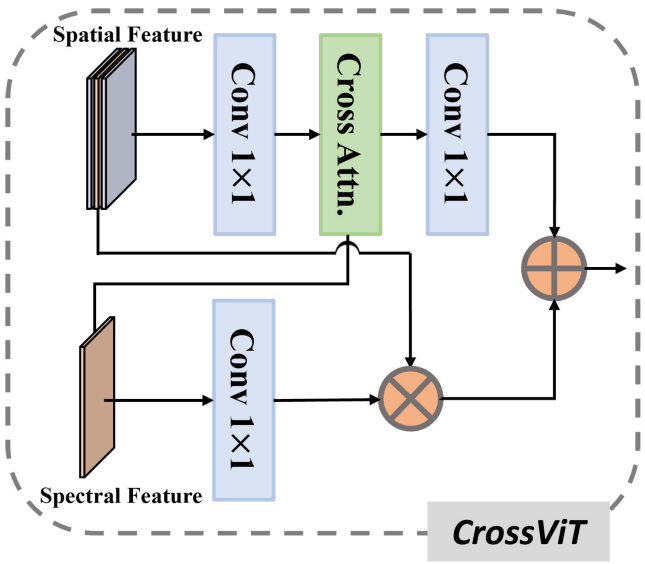


Fig. 3. Proposed CrossViT framework.

the spectral features. Consequently, we obtain two feature maps of the same size. Leveraging (7), we feed these two components into the cross-attention mechanism, with the goal of learning the representation weights of spatial features from spectral features.

For the two feature maps of the input, a fully connected layer is used to compress the size of H_{spe} to 1×512, allowing it to be elementwisely multiplied with H_{spa}. The resulting outcome is then added to the result obtained through cross-attention, thereby constructing the entire framework, i.e.,

$$F_{\text{CrossVit}}(H_{\text{spa}}, H_{\text{spe}}) = \boldsymbol{W}_1 F_C\left(\boldsymbol{W}_2 H_{\text{spa}}, H_{\text{spe}}\right) + H_{\text{spa}} \odot \boldsymbol{W}_3 H_{\text{spe}}. \tag{13}$$

It is important to note that all convolutional operations mentioned in the article are followed by a batch normalization step.

F. Classifier and Loss Function

To classify the extracted features, we add a classification head to each potential contrastive flow. The structure of this classification head is a fully connected layer with a ReLU activation and another dropout layer, which reduces the layer size to 128. We repeat this module again to further reduce the layer size to 64. Finally, a fully connected layer is employed to match the number of classes.

In our classification task, due to extracting an equal number of samples for each specific category during training, there is no issue of data imbalance in the training process. In the classification task, the loss function measures the error in predicting the category labels of the target objects. Here, we use the cross-entropy loss.

III. Experimental Results and Analysis

In this section, we introduce four well-known HSI datasets, namely Indian Pines, Pavia University, Salinas, and Houston datasets. We first discuss the current state-of-the-art methods and their implementation details as comparative approaches. Then, we conduct extensive experiments and ablation studies to quantitatively and qualitatively evaluate the performance of the proposed method for HSI classification.

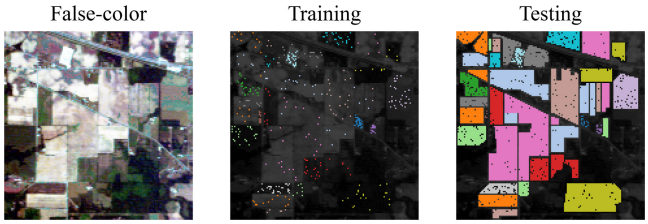


Fig. 4. False-color images and the distributions of training and testing samples for Indian Pines dataset.

TABLE I
Land-Cover Classes of the Indian Pines Dataset With the Number of Training and Testing Samples Shown for Each Class

Class No.	Class Name	Training	Testing
1	Alfalfa	30	16
2	Corn notill	30	1398
3	Corn mintill	30	800
4	Corn	30	207
5	Grass pasture	30	453
6	Grass trees	30	700
7	Grass pasture mowed	15	13
8	Hay windrowed	30	448
9	Oats	15	5
10	Soybean notill	30	942
11	Soybean mintill	30	2425
12	Soybean clean	30	563
13	Wheat	30	175
14	Woods	30	1235
15	Buildings Grass Trees Drives	30	356
16	Stone Steel Towers	30	63
-	Total	450	9799

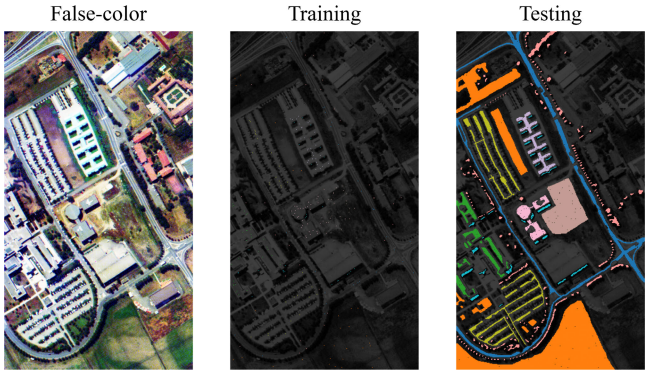


Fig. 5. False-color images and the distributions of training and testing samples for Pavia University dataset.

A. Dataset Description

1) Indian Pines Dataset: This dataset was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor located in northwest Indiana, USA. The scene contains 145×145 pixels with a ground sampling distance (GSD) of 20 m and a wavelength range of 0.4–2.5 μm. The data consists of 220 spectral bands ranging from 40 to 250 μm

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TABLE II
LAND-COVER CLASSES OF THE PAVIA UNIVERSITY DATASET WITH THE NUMBER OF TRAINING AND TESTING SAMPLES SHOWN FOR EACH CLASS

Class No.	Class Name	Training	Testing
1	Asphalt	30	6601
2	Meadows	30	18619
3	Gravel	30	2069
4	Trees	30	3034
5	Painted metal sheets	30	1315
6	Bare Soil	30	4999
7	Bitumen	30	1300
8	Self-Blocking Bricks	30	3652
9	Shadows	30	917
-	Total	270	42506

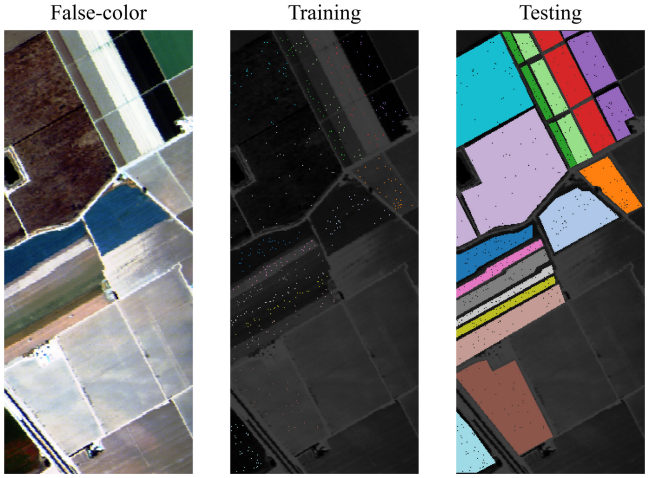


Fig. 6. False-color images and the distributions of training and testing samples for Salinas dataset.

with a spectral resolution of 10 nm. Twenty noisy and water-absorbing bands (104–108, 150–163, and 220) were removed, resulting in 200 retained bands. The data primarily represents multiple agricultural regions and includes 16 land-cover classes in total.

Table I lists the 16 major land-cover classes designed for this study scene, along with the number of training and testing samples used for the classification task. The false-color image of Indian Pines and the distribution of those samples are shown in Fig. 4.

2) Pavia University Dataset: This dataset was collected by the reflective optics system imaging spectrometer (ROSIS) sensor, corresponding to a wavelength range of 0.43–0.86 μm. During data preprocessing, 12 bands heavily affected by noise and water absorption were removed, leaving 103 spectral channels for the experiments. The dataset has a spatial resolution of 1.3 m and consists of a scene with 610 $\times$ 340 pixels, covering an area of 0.8 $\times$ 0.4 km. The scene contains nine classes in total.

Table II lists the class names along with the number of samples in the training and testing sets for this dataset. The false-color image of Pavia University and the distribution of those samples are shown in Fig. 5.

3) Salinas Dataset: This dataset has a spatial resolution of 3.7 m and covers an area of 1.9 $\times$ 0.8 km. The wavelength range is from 0.4 to 2.5 μm. During data

TABLE III
LAND-COVER CLASSES OF THE SALINAS DATASET WITH THE NUMBER OF TRAINING AND TESTING SAMPLES SHOWN FOR EACH CLASS

Class No.	Class Name	Training	Testing
1	Brocoli green weeds 1	30	1979
2	Brocoli green weeds 2	30	3696
3	Fallow	30	1946
4	Fallow rough plow	30	1364
5	Fallow smooth	30	2648
6	Stubble	30	3929
7	Celery	30	3549
8	Grapes untrained	30	11241
9	Soil vinyard develop	30	6173
10	Corn senesced green weeds	30	3248
11	Lettuce romaine 4wk	30	1038
12	Lettuce romaine 5wk	30	1897
13	Lettuce romaine 6wk	30	886
14	Lettuce romaine 7wk	30	1040
15	Vinyard untrained	30	7238
16	Vinyard vertical trellis	30	1777
-	Total	480	53649

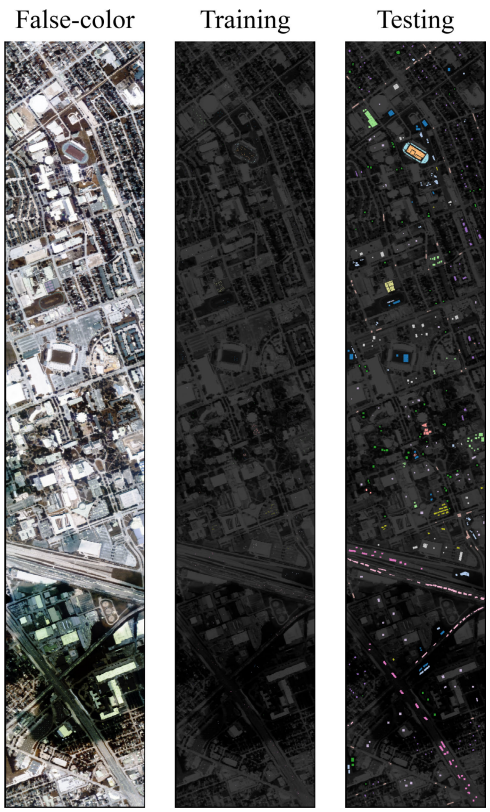


Fig. 7. False-color images and the distributions of training and testing samples for Houston 2013 dataset.

preprocessing, 20 bands heavily affected by water absorption were removed. After processing, a total of 204 spectral channels were included for the experiments. The dataset consists of 16 classes.

Table III provides the class names along with the number of samples in the training and testing sets for this dataset. The false-color image of Salinas and the distribution of those samples are shown in Fig. 6.

4) Houston 2013 Dataset: This dataset was used for the 2013 IEEE GRSS Data Fusion Contest and collected

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TABLE IV
LAND-COVER CLASSES OF THE HOUSTON 2013 DATASET WITH THE NUMBER OF TRAINING AND TESTING SAMPLES SHOWN FOR EACH CLASS

Class No.	Class Name	Training	Testing
1	Healthy Grass	30	1221
2	Stressed Grass	30	1224
3	Synthetic Grass	30	667
4	Trees	30	1214
5	Soil	30	1212
6	Water	30	295
7	Residential	30	1238
8	Commercial	30	1214
9	Road	30	1222
10	Highway	30	1197
11	Railway	30	1205
12	Parking Lot 1	30	1203
13	Parking Lot 2	30	439
14	Tennis Court	30	398
15	Running Track	30	630
-	Total	450	14579

using the ITRES CASI-1500 sensor over the campus of the University of Houston, Texas, and its surrounding rural areas. The image size is 349 × 1905 pixels. After preprocessing steps such as pose correction and radiometric calibration, the dataset was experimented upon using 144 spectral channels ranging from 364 to 1046 nm with a spectral resolution of 10 nm.

Table IV provides a list of 15 challenging land-cover categories along with the number of samples in the training and testing sets. The false-color image of Houston 2013 and the distribution of those samples are shown in Fig. 7.

B. Experimental Setup

The networks presented in this article were implemented using the PyTorch platform and optimized using the Adam optimizer. For learning rate adjustment, a step strategy was employed, dynamically reducing it by half every 30 epochs. The initial value of the learning rate is 0.0001. The maximum number of training epochs for the networks was configured at 300. Batch normalization was integrated into the process, and a batch size of 16 was selected for the training phase. All experiments were carried out using a GeForce RTX 2080 Ti graphics card.

C. Comparison With State-of-the-Art Baseline Methods

We selected several state-of-the-art baseline methods for comparison, including traditional machine learning methods such as KNN, random forest (RF), and RBF-SVM. Deep learning methods included 2D-CNN, 3D-CNN, HybridSN, FuNet, and SpectralFormer.

For the machine learning methods, we implemented them using the scikit-learn library in Python. We employed K-fold cross-validation with a value of 5 to select the optimal parameter values.

1) *KNN:* The important parameter to adjust is the number of nearest neighbors participating in the voting. The number of neighbors participating in the classification is a crucial parameter for KNN, and we dynamically selected the best parameter from the range of 1–11.
2) *RBF-SVM:* The regularization parameter C, which balances training and testing errors, is adjusted. We searched within the range of 10^{-4}–10^{6}.
3) *RF:* The parameters to adjust include the number of estimators, max depth, min sample leaf, and min samples split.

   For the first three methods, we kept the comparison consistent with the baseline methods, FuNet and SpectralFormer. We made slight adjustments to the structure design mentioned in the article to make it more reasonable.
4) *2D-CNN:* The 2D-CNN consists of three convolutional layers and two pooling layers. The first convolutional layer has a number of input channels equal to the number of spectral bands, and the output channels are set to 32. The kernel size is 3 × 3. This is followed by a 2 × 2 pooling layer and a ReLU activation function. The second convolutional layer has 64 output channels with a kernel size of 3 × 3. Similarly, it is followed by a 2 × 2 pooling layer and a ReLU activation function. The third convolutional layer has 128 output channels with a kernel size of 1 × 1, followed by a ReLU activation function. Finally, a global adaptive pooling layer is applied to reduce the feature map size to 1 × 1. Thus, the length of the output feature vector from the feature extraction network is 128. A single fully connected layer is used as the classifier. In the experiments, the patch size is set to 11.
5) *3D-CNN:* 3-D convolution is considered a structure that integrates spatial and spectral dimensions using convolutional operations. It has been widely applied in current research, and there are several relevant studies. In this article, we design three 3-D convolutional modules, which are consistent with the 3-D convolutional feature



Fig. 8. Classification maps of different methods on the Indian Pines dataset. (a) Ground truth. (b) KNN. (c) RF. (d) SVM. (e) 2D-CNN. (f) 3D-CNN. (g) HybridSN. (h) SpectralFormer. (i) DSGSF. (j) SPHGCN. (k) AMGCFN. (l) CreatingNet.

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TABLE V
CLASSIFICATION ACCURACIES (%) AND KAPPA COEFFICIENTS OF DIFFERENT MODELS ON THE INDIAN PINES DATASET. THOSE BEST ACCURACIES ARE SHOWN WITH THE BOLD TYPE FACE

Class No.	KNN	RF	SVM	2-D CNN	3-D CNN	HybridSN	SpectralFormer	DSGSF	SPHGCN	AMGCFN	proposed
1	87.50	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	48.15	**100.00**	**100.00**	**100.00**
2	42.42	26.39	34.48	28.18	54.44	72.03	37.20	87.95	90.19	76.13	**90.34**
3	48.38	48.00	0.62	54.88	66.63	85.25	27.50	85.77	**94.62**	93.61	90.75
4	64.73	67.15	67.63	**100.00**	80.19	93.72	88.41	85.04	98.54	97.58	99.03
5	79.25	79.47	60.04	77.93	77.93	83.00	56.95	81.15	**97.57**	97.34	91.17
6	86.00	82.71	60.71	96.43	79.00	99.29	72.57	99.66	97.29	**100.00**	99.86
7	92.31	92.31	0.00	**100.00**	**100.00**	**100.00**	92.31	34.15	**100.00**	**100.00**	**100.00**
8	85.71	88.62	75.45	**100.00**	99.11	100.00	90.40	98.43	99.55	95.09	100.00
9	60.00	**100.00**	0.00	0.00	**100.00**	**100.00**	**100.00**	80.00	**100.00**	**100.00**	**100.00**
10	64.86	65.82	27.28	33.02	82.17	90.45	40.23	94.78	79.34	**95.96**	95.01
11	46.47	54.47	71.71	78.43	61.24	84.54	70.02	**88.85**	75.28	88.48	86.76
12	44.94	50.44	16.70	35.88	67.32	79.04	30.20	73.99	**97.16**	93.06	89.17
13	94.29	93.71	99.43	**100.00**	**100.00**	**100.00**	**100.00**	98.80	99.43	**100.00**	**100.00**
14	76.76	84.45	51.26	99.60	84.21	96.03	73.04	99.18	98.70	98.87	**100.00**
15	39.89	55.06	1.12	91.57	93.82	96.63	53.37	93.96	99.16	**99.72**	98.03
16	98.41	98.41	92.06	**100.00**	**100.00**	**100.00**	93.65	88.52	**100.00**	**100.00**	**100.00**
OA (%)	59.15	60.73	47.32	68.93	72.40	87.26	58.17	89.79	89.34	91.90	**92.98**
AA (%)	69.49	74.19	47.41	74.74	84.13	92.50	70.37	83.34	95.43	95.99	**96.26**
κ	0.5428	0.5598	0.3899	0.6429	0.6891	0.8546	0.5177	0.8828	0.8788	0.9076	**0.9198**

TABLE VI
CLASSIFICATION ACCURACIES (%) AND KAPPA COEFFICIENTS OF DIFFERENT MODELS ON THE PAVIA UNIVERSITY DATASET. THE BEST ACCURACIES ARE SHOWN WITH THE BOLD TYPE FACE

Class No.	KNN	RF	SVM	2-D CNN	3-D CNN	HybridSN	SpectralFormer	DSGSF	SPHGCN	AMGCFN	proposed
1	70.17	64.90	52.61	57.98	68.55	84.43	55.69	95.65	**98.50**	96.20	92.23
2	81.68	81.20	50.74	92.19	92.67	93.45	83.44	98.06	95.72	91.85	**98.20**
3	78.49	0.24	18.08	56.84	59.55	82.99	40.02	85.12	98.16	**100.00**	98.36
4	76.07	88.40	95.78	96.44	97.20	98.55	88.70	98.01	81.11	97.23	**99.51**
5	99.09	99.09	99.01	99.77	94.53	99.62	88.06	97.09	99.77	**100.00**	**100.00**
6	45.87	3.08	78.28	43.57	80.20	83.04	59.93	65.42	**98.86**	96.56	95.14
7	83.31	94.46	96.85	89.54	94.77	99.15	71.23	76.96	98.15	**99.92**	99.85
8	51.89	96.28	87.98	75.78	86.25	95.18	89.43	95.16	**98.17**	91.23	97.26
9	99.89	**100.00**	**100.00**	97.27	96.51	99.35	98.26	97.74	97.60	98.26	**100.00**
OA (%)	73.55	68.72	63.06	78.59	85.82	91.32	75.23	90.58	96.05	94.44	**97.08**
AA (%)	76.27	69.74	75.48	78.82	85.58	92.86	74.97	89.91	96.23	96.81	**97.84**
κ	0.6508	0.5872	0.5520	0.7147	0.8129	0.8854	0.6760	0.8776	0.9477	0.9272	**0.9613**

extraction part of the HybridSN model mentioned later. The classification part is modified to a fully connected layer structure. The patch size is set to 25.

6) *HybridSN [28]:* HybridSN is a method that combines 2D-CNN and 3D-CNN. As a classic HSI classification model, it considers both spectral and spatial features using 3-D convolution and further enhances the features using 2-D convolution. Initially, the article uses PCA to reduce the dimensionality of the hyperspectral data to 30 dimensions. In our experiments, to construct an end-to-end model and enable the linear combinations between bands to be learnable, we replace the PCA method with 1×1 2-D convolutional kernels. The output channel of the initial dimension reduction convolutional layer is changed to 30 to maintain consistency with the subsequent structure.
7) *SpectralFormer [31]:* This method is the first to fully incorporate ViTs into HSI classification tasks. In this article, we adopt the patch-wise version of SpectralFormer as the comparative method.
8) *DSGSF [24]:* Dual-view spectral and global spatial feature fusion network using the global spatial representation and the dual-view spectral information to capture the discriminative features of HSI.
9) *SPHGCN [23]:* Structure preserved hypergraph convolution network first builds a multiple neighborhood reconstruction model to reveal the essential resemblance of multiple pixels in nonlinear spectral feature space.
10) *AMGCFN [22]:* Attention Multi-hop Graph and Multiscale Convolutional Fusion Network makes full use of multiscale convolution and multihop graph features, which is conducive to the learning of multilevel contextual semantic features.

D. Experimental Result and Visual Comparison

To assess the efficacy of our proposed methods, this study conducts a comparative analysis against a range of benchmarks. These benchmarks encompass three conventional machine learning algorithms, i.e., KNN, RBF-SVM, and RF.

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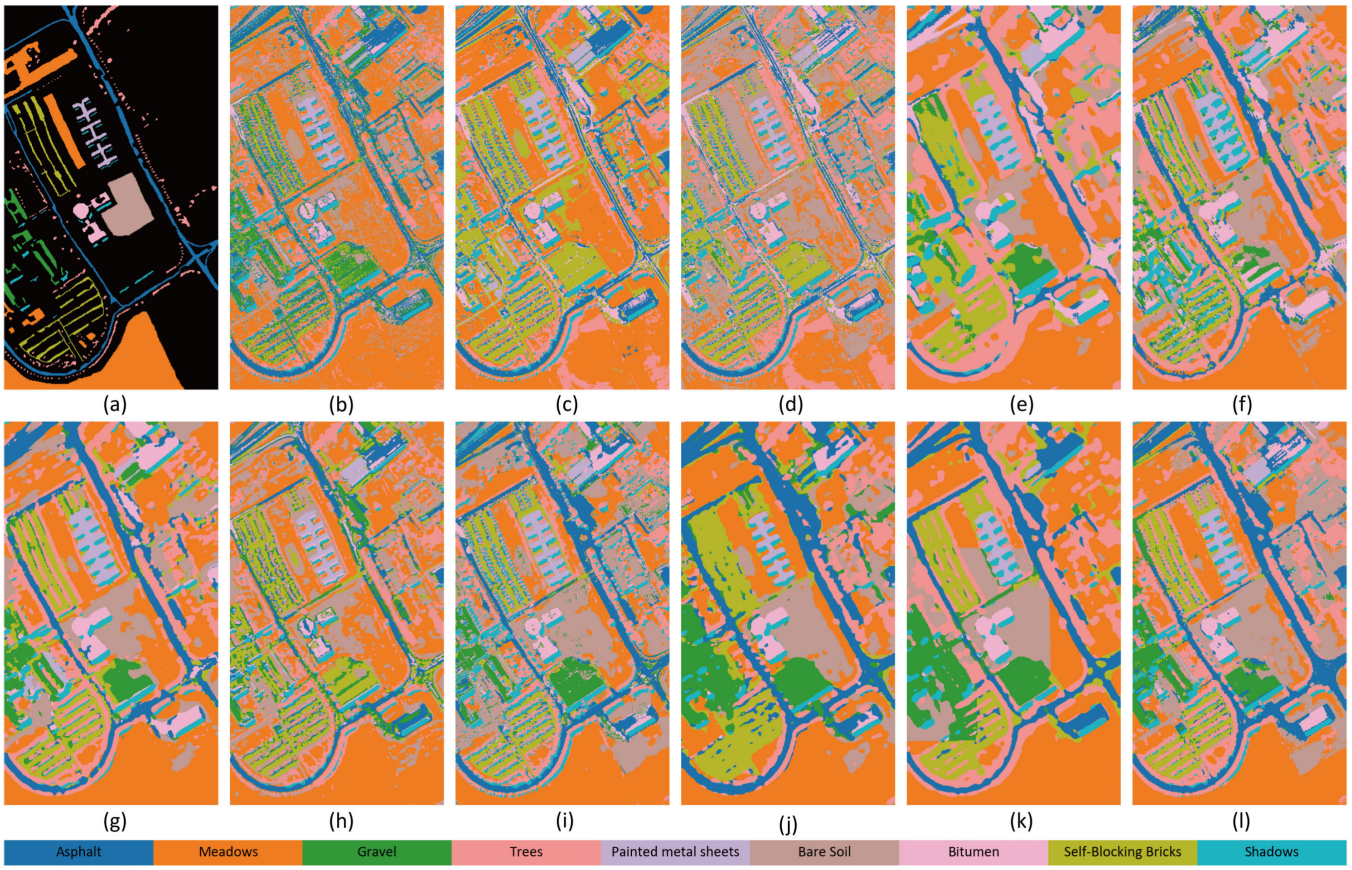


Fig. 9. Classification maps of different methods on the Pavia University dataset. (a) Ground truth. (b) KNN. (c) RF. (d) SVM. (e) 2D-CNN. (f) 3D-CNN. (g) HybridSN. (h) SpectralFormer. (i) DSGSF. (j) SPHGCN. (k) AMGCFN. (l) CreatingNet.

TABLE VII

CLASSIFICATION ACCURACIES (%) AND KAPPA COEFFICIENTS OF DIFFERENT MODELS ON THE SALINAS DATASET. THOSE BEST ACCURACIES ARE SHOWN WITH THE BOLD TYPE FACE

Class No.	KNN	RF	SVM	2-D CNN	3-D CNN	HybridSN	SpectralFormer	DSGSF	SPHGCN	AMGCFN	proposed
1	97.37	99.29	98.48	**100.00**	84.64	99.80	91.36	89.12	**100.00**	99.80	99.95
2	98.76	99.11	83.04	97.19	81.12	99.92	99.43	99.77	99.78	99.73	**100.00**
3	97.43	90.90	52.26	74.92	98.25	**100.00**	92.50	96.36	99.54	**100.00**	**100.00**
4	99.05	98.17	98.53	**100.00**	99.41	99.85	97.95	95.00	99.93	**100.00**	99.49
5	93.88	96.34	97.73	91.78	**100.00**	**100.00**	94.00	99.45	98.68	94.07	99.43
6	99.57	99.47	98.17	**100.00**	99.98	**100.00**	99.92	99.92	98.32	99.87	99.95
7	98.25	98.25	99.24	99.94	90.84	97.58	97.46	96.90	98.67	99.55	**99.89**
8	68.06	58.94	56.71	78.36	70.29	75.51	73.53	88.80	82.72	81.98	**90.04**
9	93.73	98.98	94.33	97.26	92.03	99.84	90.60	99.50	99.90	99.16	**100.00**
10	87.78	84.94	7.27	93.41	92.46	98.43	78.97	92.75	93.99	95.97	**99.60**
11	92.58	90.17	73.41	99.62	97.69	98.75	95.95	97.81	98.65	99.71	**100.00**
12	99.63	98.05	61.10	85.98	**100.00**	99.95	99.32	99.89	**100.00**	**100.00**	**100.00**
13	98.08	98.42	99.10	**100.00**	97.97	**100.00**	99.55	98.19	99.77	**100.00**	99.32
14	89.71	88.65	86.25	99.04	97.12	99.52	95.58	92.46	99.04	**100.00**	96.06
15	53.30	65.61	67.52	71.08	73.67	87.61	79.50	72.49	94.58	**95.55**	92.04
16	88.24	94.82	61.11	96.40	87.96	99.83	97.41	98.84	97.64	**100.00**	99.44
OA (%)	84.02	84.26	73.53	88.70	85.76	92.87	87.90	91.90	94.84	94.92	**96.65**
AA (%)	90.96	91.26	77.14	92.81	91.46	97.29	92.69	94.83	97.58	97.84	**98.45**
κ	0.8224	0.8255	0.7062	0.8743	0.8417	0.9207	0.8658	0.9100	0.9427	0.9436	**0.9627**

Additionally, we compare our methods to three fundamental network architectures: 2D-CNN and 3D-CNN. Furthermore, we evaluate their performance relative to five classic HSI classification methods, i.e., HybridSN, SpectralFormer, DSGSF, SPHGCN, and AMGCFN. The parameter settings for each method draw heavily from their recommended configurations and prior knowledge in the field.

In Tables V–VIII, among all the experiments on the four datasets, the three worst-performing machine learning methods are KNN, RF, and SVM. KNN, RF, and SVM, which

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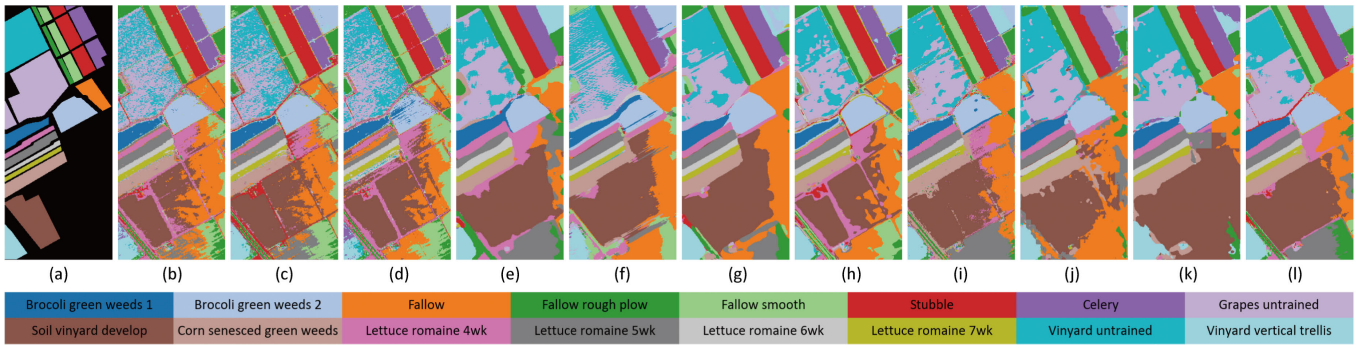


Fig. 10. Classification maps of different methods on the Salinas dataset. (a) Ground truth. (b) KNN. (c) Random forest. (d) SVM. (e) 2D-CNN. (f) 3D-CNN. (g) HybridSN. (h) SpectralFormer. (i) DSGSF. (j) SPHGCN. (k) AMGCFN. (l) CreatingNet.

TABLE VIII

CLASSIFICATION ACCURACIES (%) AND KAPPA COEFFICIENTS OF DIFFERENT MODELS ON THE HOUSTON 2013 DATASET. THOSE BEST ACCURACIES ARE SHOWN WITH THE BOLD TYPE FACE

Class No.	KNN	RF	SVM	2-D CNN	3-D CNN	HybridSN	SpectralFormer	DSGSF	SPHGCN	AMGCFN	proposed
1	97.05	97.46	96.81	84.03	88.94	98.77	**100.00**	95.33	91.89	95.33	95.25
2	98.28	91.99	85.62	57.44	69.94	96.41	87.58	95.48	89.94	89.37	**99.75**
3	97.75	92.35	98.65	92.05	**100.00**	99.85	87.71	**100.00**	98.95	**100.00**	**100.00**
4	93.90	91.60	90.94	92.26	81.14	91.85	90.77	99.73	91.67	98.19	**99.84**
5	95.63	94.55	92.74	**100.00**	86.22	**100.00**	97.03	98.91	94.72	**100.00**	**100.00**
6	91.19	83.73	82.71	86.78	84.75	87.46	76.61	91.35	91.16	91.53	**95.25**
7	75.93	73.02	29.56	78.19	81.91	92.57	72.54	99.37	96.35	**98.87**	96.69
8	62.44	54.37	16.31	56.59	59.56	68.78	71.01	87.86	86.69	73.87	**90.94**
9	68.17	64.65	80.03	38.63	75.94	88.79	72.18	91.01	86.33	**93.04**	91.65
10	76.61	51.80	26.48	44.19	80.03	94.99	70.34	85.84	96.57	95.31	**98.16**
11	71.37	79.59	61.08	71.45	80.75	90.71	60.50	87.98	96.01	**99.42**	96.35
12	58.60	50.12	26.27	39.15	67.66	80.88	64.67	83.82	87.70	92.01	**96.01**
13	31.89	41.69	10.48	95.67	87.47	94.08	67.65	93.95	94.74	**95.43**	90.21
14	90.95	91.46	96.48	99.75	94.72	**100.00**	96.23	86.77	**100.00**	**100.00**	99.75
15	98.73	95.24	98.89	**100.00**	95.08	**100.00**	98.73	98.24	**100.00**	99.37	97.78
OA (%)	80.56	76.27	63.96	71.10	79.97	91.55	80.08	92.86	92.78	94.28	**96.56**
AA (%)	80.57	76.91	66.20	75.75	82.27	92.34	80.90	93.04	93.53	94.78	**96.51**
κ	0.7898	0.7436	0.6114	0.6880	0.7836	0.9086	0.7848	0.9228	0.9220	0.9381	**0.9627**

primarily rely on spectral features without integrating spatial information via patch processing, exhibit limited classification performance. Among these three methods, SVM performs the least effectively due to its strong sensitivity to the limited size of the training dataset, resulting in considerable disparities in classification accuracy across different classes. Remarkably, the KNN method displays relatively strong performance, indicative of robust spectral correlations within the same class, rendering it a feasible approach. In contrast, RF consistently demonstrates stable performance, attributed to its capacity to model spectral data in a more intricate and profound manner using numerous decision trees. Notably, our method surpasses the others in both stability and classification accuracy.

In the realm of deep learning methods, there is less variation in classification accuracy among different classes in the dataset compared to traditional methods. This indicates that deep learning methods tend to perform relatively well in classes that are challenging for traditional methods. However, for basic CNN networks, there is not always a well-established mapping for some of the more easily distinguishable classes. In general, deep learning methods demonstrate a substantial enhancement over traditional methods, even when employing straightforward 2-D and 3-D convolutional networks.

2-D convolutional methods effectively extract spatial features from images. On the other hand, 3-D convolutions can simultaneously capture both spatial and spectral features from images. Due to their feature richness, 3-D convolutions often yield more substantial improvements compared to 2-D convolutions. The HybridSN method, which combines both types of convolutional kernels, benefits from the advantages of each, resulting in stable and significant enhancements.

SpectralFormer is esteemed in the field and have introduced novel adjustments and breakthroughs in network structure. Nevertheless, its performance across different datasets tends to be inconsistent. In contrast, our method consistently demonstrates significant improvements when compared to various other methods.

DSGSF, SPHGCN, and AMGCFN are all patch-free methods that demonstrate significant performance advantages compared to general deep learning methods. However, during actual training, these methods tend to favor smaller subsets of the entire HSI. When dealing with larger datasets, they can significantly increase computation time and memory usage,

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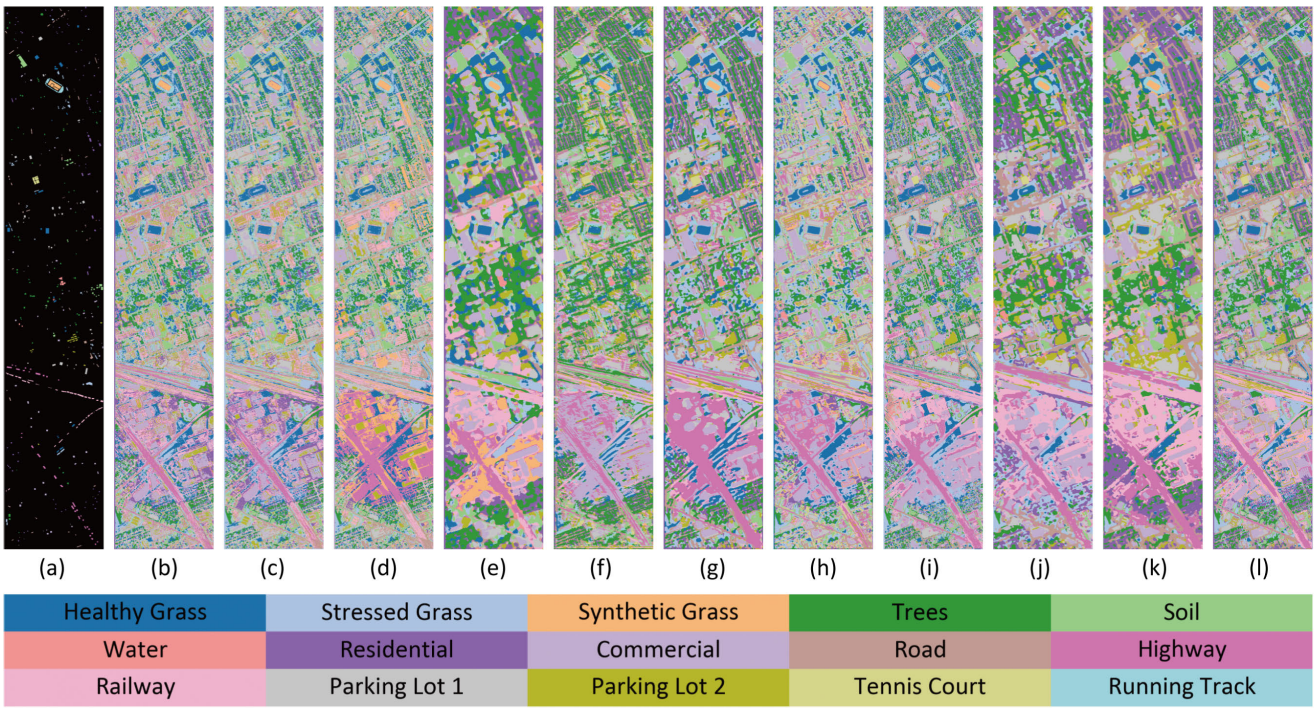


Fig. 11. Classification maps of different methods on the Houston 2013 dataset. (a) Ground truth. (b) KNN. (c) RF. (d) SVM. (e) 2D-CNN. (f) 3D-CNN. (g) HybridSN. (h) SpectralFormer. (i) DSGSF. (j) SPHGCN. (k) AMGCFN. (l) CreatingNet.

thereby limiting their performance. In contrast, the patch-based approach can overcome the limitations imposed by the size of the entire image, making it more versatile and practical for real-world applications.

To visually showcase the effectiveness of our approach, we present classification result images for all the methods discussed in Figs. 8–11. Spectral-based methods exhibit a considerable amount of salt-and-pepper noise, validating the lower classification accuracy reported in the table due to their disregard for spatial information. Conversely, large patch-based CNN methods produce overly smoothed boundaries, resulting in the loss of finer details in the resulting images. This is attributed to the fact that, with large-scale patches, neighboring pixels often exhibit high overall similarity, diminishing the prominence of the queried pixel at the center.

However, in comparison to other methods, the classification maps generated by our proposed approach are notably more precise, smoother in terms of spatial consistency, and contain fewer inaccurately labeled pixels, which aligns with the results presented in the table. Notably, in the Houston 2013 dataset, where certain sections are concealed by cloud cover and the lower portion of the road is frequently misclassified with unclear boundaries. Due to the effective spectral modeling capability of our method, which aids the discriminator in achieving better discrimination, it provides clearer boundary delineation for general patch-based methods. This also highlights the effective enhancement that cross semantic modeling brings to traditional deep learning methods.

E. Discussion and Analysis

1) Ablation Study: Moreover, to underscore the effectiveness of our proposed method, we conducted a comprehensive comparison with three alternative approaches, i.e., a singular stream dedicated to spatial feature extraction, a singular stream for spectral feature extraction, and a fusion strategy that amalgamates the two feature streams without CrossViT. The patch size remained consistent based on the insights gained from previous experiments. The obtained experimental results are shown in Table IX. This comparative analysis unveiled that the solitary spatial feature extraction stream is notably sensitive to specific features, while the solitary spectral feature extraction stream displays sensitivity to distinct classes. When these streams are fused, the synergy effectively bridges the gap, complementing information from both dimensions, thus further enhancing classification performance. Notably, our proposed feature fusion strategy outperforms relatively simpler fusion methods, confirming the performance improvement.

2) Parameters Analysis: In assessing the efficacy of the input modules, we performed a systematic grid test to evaluate the input sizes for the spatial and spectral streams. The obtained experimental results are shown in Table X. The outcomes indicate that optimizing the input sizes for each stream to align with their respective network structures yields the best performance. This personalized approach to input sizes results in enhanced network performance.

3) Different Numbers of Training Samples: Pixel labeling of HSIs is costly, resulting in a limited number of training samples. In previous experiments, we have demonstrated that CreatingNet performs exceptionally well with small sample sizes. To assess the stability and robustness of our method across different training sample sizes, we ran it 10 times on the Indian Pines dataset at intervals of 20% [20%, 40%, 60%, 80%, 100%]. The average OA results and standard deviations of the CreatingNet proposed in this study are shown in Fig. 12.

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TABLE IX
CLASSIFICATION ACCURACY OA(%) (KAPPA COEFFICIENTS) OF THE ABLATION EXPERIMENTS FOR CREATINGNET ON THE FOUR DATASETS

	Indian Pines	Pavia University	Salinas	Houston 2013
only spatial	86.59 (0.8477)	95.40 (0.9390)	92.60 (0.9178)	92.47 (0.9185)
only spectral	85.84 (0.8106)	86.92 (0.8990)	88.30 (0.8578)	91.97 (0.9132)
without CrossViT	87.14 (0.8520)	94.93 (0.9333)	90.13 (0.8903)	93.49 (0.9238)
proposed	**92.98 (0.9198)**	**97.08 (0.9613)**	**96.65 (0.9627)**	**96.56 (0.9627)**

TABLE X
IMPACT OF DIFFERENT INPUT PLATE SIZES ON MODEL CLASSIFICATION ACCURACY (%) ON THE HOUSTON 2013 DATASET

w_{spa} \ w_{spe}	3	5	7
7	91.47	90.31	87.60
9	93.16	91.42	90.41
11	95.17	92.70	91.74
13	**96.56**	93.85	93.98
15	96.03	94.76	95.23
17	95.75	94.59	94.43

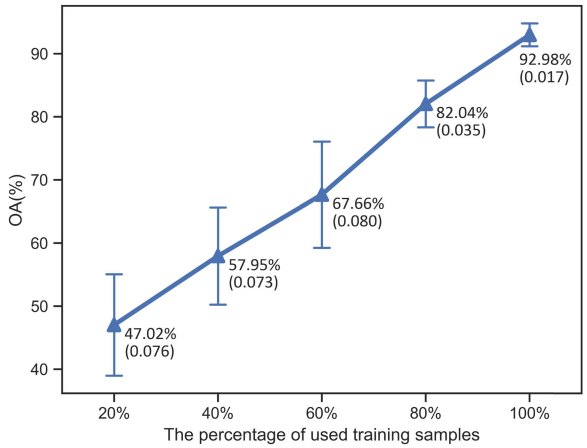


Fig. 12. Classification results (OA) obtained by our proposed CreatingNet with the varying number of used training samples on the Indian Pines dataset.

The OA results exhibit a reasonable trend: as the percentage of training samples increases, the classification performance gradually improves. When more training samples are included, OA stabilizes, which strongly indicates the stability of the proposed method.

IV. CONCLUSION

Leveraging the wealth of spatial and spectral information in HSI classification, this article introduces an efficient multiscale fusion network structure designed for spectral–spatial feature extraction. Our model harnesses a residual network structure to adequately extract spatial features, while simultaneously employing an MSA mechanism to extract spectral features. The synergistic fusion of both feature representations is achieved through a multiscale feature fusion module. To seamlessly align with the dual-stream network structure, we tailor patch sizes for distinct network architectures based on comprehensive theoretical analyses. The efficacy of the proposed modules is corroborated through ablation experiments. A comparative evaluation against state-of-the-art methods across four diverse scene datasets underscores the superior performance of our proposed method, showcasing its exceptional effectiveness. In future work, we will focus on optimizing the network structure and promoting deeper integration of features to improve model training efficiency and classification accuracy.

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